Constructing Analytical Solutions for Two-Dimensional Two-Group Neutron Diffusion Equations via Evolutionary Program Synthesis

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Abstract

Finite difference and finite element solvers for neutron diffusion accumulate discretization error; physics-informed neural networks avoid the mesh but require lengthy training. We take a different route: a Fourier cosine expansion in the transverse direction decouples the two-group neutron diffusion equations into per-mode ODEs whose characteristic equations yield closed-form fast and thermal flux solutions. FaMoU, our LLM-driven evolutionary program synthesis engine, then seeds from this analytical solution and iteratively mutates the solver code, scored by an automated physics evaluator. On a standardized benchmark, the Fourier analytical solution (N=30 modes) achieves a combined PDE+BC score of 0.9989, reducing the combined residual 39-fold relative to PINN and surpassing both FDM variants under uniform pointwise evaluation. An overflow-safe adaptive variant (up to N=221 modes) reaches 0.9997. FaMoU seeds from this solution; after 10 iterations the LLM-evolved program already surpasses the seed, confirming that evolutionary program synthesis is a viable path for physics PDE discovery.

1 Introduction

Nuclear reactor design and safety analysis depend on accurate neutron flux predictions. The neutron diffusion equation—a tractable approximation to the Boltzmann transport equation—remains the standard model for core calculations, and its two-group variant, separating fast and thermal neutrons, retains enough physics to capture moderation and absorption while admitting analytical treatment in simple geometries [1–3].

Solutions to the multigroup diffusion equations have traditionally come from two directions. Analytical methods—Fourier series, Green's functions, eigenvalue decomposition—yield exact results for canonical geometries and remain the primary verification benchmarks [4, 5], but they are restricted to regular domains with simple material layouts; deriving them for coupled multigroup systems under non-trivial boundary conditions requires considerable manual effort. Numerical discretizations (FDM, FEM) handle arbitrary geometries and heterogeneous media at the price of discretization error and, for full-core problems, substantial computation [3, 6].

Scientific machine learning has opened a third route. Physics-Informed Neural Networks (PINNs) fold PDE residuals into the training loss, producing mesh-free solutions for both forward and inverse problems [7]. In reactor physics, successive architectural changes have reduced PINN errors on two-group problems [8–10]. Neural operator approaches, including

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DeepONet [11] and the Fourier Neural Operator [12], learn solution operators that generalize across problem parameters, achieving large speedups over traditional solvers in neutron transport surrogate modeling [13]. Both families have limits: PINNs exhibit training instability and accuracy bottlenecks (0.6–15% error for two-group problems [8]), and neural operators are constrained by the coverage and fidelity of their training data.

This paper combines classical analytical derivation with LLM-driven program evolution to solve the two-group neutron diffusion equation on a two-dimensional finite homogeneous domain with boundary sources. The first component is a Fourier cosine series solution that gives exact ground truth for the boundary-source problem in rectangular geometry. The second is **FaMoU** (Fast Mode mUtation), an evolutionary program synthesis loop—inspired by FunSearch and AlphaEvolve—in which a large language model iteratively generates and refines solver programs, scored against the analytical ground truth. We benchmark both components alongside FDM, high-order FDM, and PINN on the same problem instance.

Specifically, we derive the two-group neutron flux distribution by expanding the solution in a Fourier cosine series that satisfies the homogeneous Neumann boundary conditions, with the boundary source incorporated through Fourier decomposition of the inhomogeneous condition. The resulting coupled system of ordinary differential equations is solved analytically for each Fourier mode, yielding pointwise ground truth with controllable accuracy governed by the number of retained modes. For the evolutionary component, inspired by FunSearch [14] and AlphaEvolve [15], an initial seed program is iteratively improved by a large language model, with each candidate scored by an automated evaluator that measures PDE and boundary-condition residuals as RMS values at fixed test points. A distinguishing feature of our setup is that the evaluator computes error against an *exact* analytical ground truth, rather than relying on heuristic scores or combinatorial bounds as in most program evolution work.

Our main contributions are as follows:

- **Analytical benchmark.** We derive and implement the complete analytical solution for the two-group neutron diffusion equation with boundary sources in a 2D finite homogeneous rectangular medium, providing a high-precision reference solution that satisfies the PDE exactly in the continuous limit and achieves residuals below 4×10^{-5} under numerical evaluation.

- **First LLM-driven evolution for physics PDE solving.** To our knowledge, this is the first work to apply the LLM-driven program evolution paradigm—pioneered by FunSearch [14] and AlphaEvolve [15] for mathematical and algorithmic discovery—to the domain of computational physics PDE solving. We demonstrate that evolutionary program synthesis can discover solver programs achieving competitive accuracy with hand-crafted numerical methods.

- **Comprehensive method comparison.** We present a unified benchmark comparing analytical, FDM, PINN, and LLM-evolved solvers on the same problem instance, providing quantitative insights into the accuracy–efficiency trade-offs across four distinct methodological paradigms.

- **Overflow-safe adaptive truncation.** We identify and resolve a numerical overflow issue in the standard Fourier implementation, deriving a stability criterion that allows the expansion to include all float64-safe modes (up to n=221), improving the combined score from 0.9989 to 0.9997.

The remainder of this paper is organized as follows. Section 2 surveys related work across classical numerical methods, PINNs, neural operators, and LLM-driven program evolution. Section 3 presents the mathematical formulation and analytical solution. Section 4 describes the LLM-driven evolutionary framework and baseline methods. Section 5 reports experimental results and comparative analysis. Section 6 concludes with a discussion of implications and future directions.

2 Related Work

2.1 Classical Numerical Methods for Neutron Diffusion

The neutron diffusion equation has been studied extensively since the early development of nuclear reactor theory. The foundational texts by Duderstadt and Hamilton [1] and Bell and Glasstone [2] established the multigroup diffusion framework, providing analytical solutions for canonical geometries (slab, cylinder, sphere) and laying the theoretical groundwork for numerical implementations.

Analytical methods. Fourier series expansion has been a classical tool for solving the diffusion equation in rectangular geometries. The finite Fourier sine transform was applied to two-dimensional neutron diffusion as early as the 1970s [16], with subsequent work by Williams et al. [5] extending Fourier methods to rectangular lattice cells with one- and two-group formulations. More recently, Zanette, Ceolin, and collaborators developed the Fictitious Borders Power Method (FBPM) [4], which applies the finite Fourier sine transform at each iteration of the power method, yielding analytical solutions for multigroup diffusion in 2D Cartesian geometry with competitive runtime compared to purely numerical approaches. For spherical geometries, Momani et al. [17] applied the homotopy perturbation method to two-energy-group models with fractional-order extensions.

Finite difference and finite element methods. The finite difference method (FDM) remains the most widely used numerical approach for neutron diffusion, with second-order central difference schemes providing $O(h^2)$ accuracy on structured grids [3, 6]. Modern implementations employ Krylov subspace methods for the resulting large sparse systems, enabling efficient multi-dimensional, multigroup calculations. The finite element method (FEM) offers greater geometric flexibility, with Lagrangian elements of varying order ($p = 1, 2, 3$) applied to two-dimensional diffusion problems [18]. Hybrid mixed finite elements have been employed for coupled multigroup systems treated as elliptic–parabolic PDEs [19]. While both FDM and FEM are mature and well-validated, they introduce discretization errors that require careful mesh convergence studies and cannot provide closed-form reference solutions.

2.2 Physics-Informed Neural Networks

Physics-Informed Neural Networks (PINNs), introduced by Raissi et al. [7], embed governing equations directly into the training loss, enabling mesh-free PDE solving. The original framework demonstrated data-driven solution and discovery of nonlinear PDEs using fully connected networks trained with automatic differentiation to enforce PDE residuals at collocation points.

Applications to nuclear reactor physics. Elhareef and Wu [8] presented the first systematic study applying PINNs to nuclear reactor calculations, addressing both fixed-source and k-eigenvalue problems. Their results revealed a significant accuracy range: 0.63% error for simple fixed-source configurations but up to $\sim$15% for two-group eigenvalue problems, with k-eigenvalue deviations of 0.13–0.92%. These findings highlight the challenge of applying PINNs to coupled multigroup systems. Subsequent work has addressed specific PINN limitations in the nuclear context. The Transfer Learning PINN (TL-PINN) approach [9] achieved up to two orders of magnitude acceleration in training for reactor transient prediction by leveraging pre-trained models across related configurations. Parameterized PINNs with hard boundary constraints [20] demonstrated 1,000$\times$ speedup over the commercial solver COMSOL for parametric steady-state diffusion problems. More recently, NAS-PINN [10] employed neural architecture search to dynamically optimize the

PINN structure for different reactor geometries, while the R^2-PINN [21] replaced the standard fully connected architecture with convolutional networks featuring skip connections to address gradient vanishing in deeper networks.

Key trends. The trajectory of PINN research in nuclear physics reveals a progression from simple architectures toward specialized designs: architectural innovations (FCN $\rightarrow$ CNN with skip connections $\rightarrow$ NAS-optimized), reusability mechanisms (transfer learning, parameterization), and adaptive training strategies (residual resampling, hard boundary constraints). Despite these advances, PINNs for multigroup diffusion remain limited by training instability and accuracy ceilings, particularly for coupled systems.

2.3 Neural Operator Approaches

Neural operators learn mappings between function spaces, offering a fundamentally different approach from PINNs by learning entire solution operators rather than individual solutions.

DeepONet. Lu et al. [11] introduced the Deep Operator Network (DeepONet), consisting of a branch network encoding the input function and a trunk network encoding the output domain coordinates. Based on the universal approximation theorem for operators, DeepONet has been applied to nuclear energy systems for real-time digital twin inference, where it was used to construct surrogate models for neutron flux spatial distributions with performance exceeding that of standard CNN and FCN architectures [13].

Fourier Neural Operator. The Fourier Neural Operator (FNO), proposed by Li et al. [12], parameterizes integral kernels directly in Fourier space, achieving state-of-the-art accuracy among learning-based PDE solvers with up to three orders of magnitude speedup over traditional methods. The FNO was the first ML-based method to achieve zero-shot super-resolution for turbulent flow simulations. A recent comparative study on neutron transport [22] found that FNO generally achieves higher predictive accuracy ($\sim$112 pcm deviation) while DeepONet offers greater computational efficiency, with both reducing runtime to less than 0.1% of a traditional S_N solver.

Relationship to our work. While neural operator approaches achieve impressive speedups, they are primarily *surrogate models* that approximate existing solvers. In contrast, our LLM-driven evolutionary approach aims to *discover* new solver programs that can be inspected, interpreted, and potentially provide insights into the problem structure.

2.4 LLM-Driven Program Evolution

The use of large language models for scientific discovery through program synthesis has emerged as an active area with rapid recent progress.

FunSearch. Romera-Paredes et al. [14] introduced FunSearch, an evolutionary system that pairs a pre-trained LLM (as a code generator) with an automated evaluator. Operating as a distributed system with a programs database employing the island model for diversity maintenance, FunSearch iteratively generates, evaluates, and selects programs in an evolutionary loop. FunSearch achieved the first genuine LLM-driven mathematical discovery, producing new constructions for the cap set problem—the largest improvement in asymptotic lower bounds in 20 years—and discovering more effective bin-packing heuristics. Critically, FunSearch searches in *function space* rather than solution space, yielding interpretable programs that describe *how to solve* a problem rather than just *what the solution is*.

AlphaEvolve. Building on FunSearch, AlphaEvolve [15] extended the paradigm to evolve entire codebases rather than single functions, using an ensemble of Gemini models for balanced throughput and solution quality. AlphaEvolve achieved remarkable results: improving Strassen's matrix multiplication algorithm (the first improvement in 56 years), recovering 0.7% of Google's worldwide compute resources through data center optimization, and rediscovering state-of-the-art solutions for 75% of 50+ open mathematical problems while finding superior solutions for 20%. Open-source implementations, such as CodeEvolve [23], have further democratized this paradigm.

Gap: LLM evolution for physics PDE solving. Existing LLM-driven program evolution work has focused primarily on combinatorial optimization, pure mathematics, and computational infrastructure. To our knowledge, no prior work has applied this paradigm to the domain of physics PDE solving. Our work addresses this gap by combining LLM-driven program evolution with an analytical ground truth evaluator, enabling the evolutionary discovery of PDE solver programs for neutron diffusion equations. Our work connects these two threads: LLM-driven program evolution, operating against an exact analytical ground truth evaluator, discovers PDE solver programs for neutron diffusion in a domain where no prior evolutionary approach has been applied.

3 Problem Formulation

We consider the steady-state two-group neutron diffusion problem on a square domain. The goal is to construct analytical functions $\varphi_1(x, y)$ and $\varphi_2(x, y)$ that simultaneously satisfy the governing PDEs and all boundary conditions.

3.1 Governing Equations

The two-group neutron diffusion equations couple the fast-group flux φ_1 and the thermal-group flux φ_2 as follows.

Fast-group equation.

$$-D_1 \nabla^2 \varphi_1 + \Sigma_r \, \varphi_1 = \nu \Sigma_{f1} \, \varphi_1 + \nu \Sigma_{f2} \, \varphi_2, \tag{1}$$

Thermal-group equation.

$$-D_2 \nabla^2 \varphi_2 + \Sigma_{a2} \, \varphi_2 = \Sigma_{1 \to 2} \, \varphi_1, \tag{2}$$

where the Laplacian operator is defined as $\nabla^2 \varphi = \frac{\partial^2 \varphi}{\partial x^2} + \frac{\partial^2 \varphi}{\partial y^2}$.

3.2 Physical Parameters

Table 1 summarizes the physical constants used throughout this work.

3.3 Domain and Boundary Conditions

The spatial domain is a unit square centered at the origin:

$$\Omega = [-0.5, \, 0.5]^2 \subset \mathbb{R}^2. \tag{3}$$

All four boundaries carry Neumann (current) conditions. Specifically, the left boundary carries a non-zero source, while the remaining three boundaries are zero-flux:

Table 1: Physical parameters for the two-group neutron diffusion problem.

Symbol	Description	Value
D_1	Fast-group diffusion coefficient	1.0
D_2	Thermal-group diffusion coefficient	0.5
Σ_r	Fast-group removal cross-section	0.02
Σ_{a2}	Thermal-group absorption cross-section	0.1
ν	Average neutrons per fission	2.5
Σ_{f1}	Fast-group fission cross-section	0.005
Σ_{f2}	Thermal-group fission cross-section	0.1
$\Sigma_{1\to 2}$	Group transfer cross-section	0.015

Left boundary ($x = -0.5$, non-zero Neumann).

$$-D_1 \frac{\partial \varphi_1}{\partial x}\bigg|_{x=-0.5} = y, \qquad -D_2 \frac{\partial \varphi_2}{\partial x}\bigg|_{x=-0.5} = y. \tag{4}$$

Right boundary ($x = 0.5$, zero Neumann).

$$-D_1 \frac{\partial \varphi_1}{\partial x}\bigg|_{x=0.5} = 0, \qquad -D_2 \frac{\partial \varphi_2}{\partial x}\bigg|_{x=0.5} = 0. \tag{5}$$

Top boundary ($y = 0.5$, zero Neumann).

$$-D_1 \frac{\partial \varphi_1}{\partial y}\bigg|_{y=0.5} = 0, \qquad -D_2 \frac{\partial \varphi_2}{\partial y}\bigg|_{y=0.5} = 0. \tag{6}$$

Bottom boundary ($y = -0.5$, zero Neumann).

$$-D_1 \frac{\partial \varphi_1}{\partial y}\bigg|_{y=-0.5} = 0, \qquad -D_2 \frac{\partial \varphi_2}{\partial y}\bigg|_{y=-0.5} = 0. \tag{7}$$

3.4 Evaluation Criteria

A candidate solution (φ_1, φ_2) is assessed by its PDE residuals and boundary condition residuals.

PDE residuals. At each interior test point $(x, y) \in \Omega$, we compute:

$$\mathrm{Res}_1(x, y) = -D_1 \left(\frac{\partial^2 \varphi_1}{\partial x^2} + \frac{\partial^2 \varphi_1}{\partial y^2} \right) + \Sigma_r \, \varphi_1 - \nu \Sigma_{f1} \, \varphi_1 - \nu \Sigma_{f2} \, \varphi_2, \tag{8}$$

$$\mathrm{Res}_2(x, y) = -D_2 \left(\frac{\partial^2 \varphi_2}{\partial x^2} + \frac{\partial^2 \varphi_2}{\partial y^2} \right) + \Sigma_{a2} \, \varphi_2 - \Sigma_{1\to 2} \, \varphi_1. \tag{9}$$

The test points used for verification are:

$$\mathcal{T} = \{(0, 0),\ (0.2, 0.2),\ (-0.2, -0.3),\ (0.4, -0.4)\}. \tag{10}$$

Boundary condition residuals. For each boundary segment Γ_i, we evaluate the deviation from the prescribed Neumann data. The combined score aggregates both PDE and BC residuals as root-mean-square (RMS) values.

Combined score. The overall quality metric is defined as:

$$S = \frac{1}{1 + \bar{R}_{\mathrm{PDE}} + \bar{R}_{\mathrm{BC}}}, \tag{11}$$

where $\bar{R}_{\mathrm{PDE}}$ and $\bar{R}_{\mathrm{BC}}$ denote the root-mean-square (RMS) residuals over the interior test points and boundary evaluation points, respectively. The score $S \in (0, 1]$, with $S = 1$ indicating a perfect solution. Programs that fail to execute or produce invalid outputs receive $S = 0$.

4 Method

This section presents the mathematical formulation of the benchmark problem and the analytical solution procedure used to generate the reference solution for the FaMoU evolutionary framework.

4.1 Two-Group Neutron Diffusion Equations

We consider the steady-state two-group neutron diffusion model on the square domain $\Omega = [-0.5,\, 0.5]^2$:

$$-D_1 \nabla^2 \varphi_1 + \Sigma_r \varphi_1 = \nu \Sigma_{f1}\, \varphi_1 + \nu \Sigma_{f2}\, \varphi_2, \tag{12}$$

$$-D_2 \nabla^2 \varphi_2 + \Sigma_{a2}\, \varphi_2 = \Sigma_{1\to 2}\, \varphi_1, \tag{13}$$

where $\varphi_1(x, y)$ and $\varphi_2(x, y)$ denote the fast and thermal neutron fluxes, respectively, with physical parameters as specified in Table 1.

The boundary conditions impose an inhomogeneous incoming current on the left face and reflective (zero Neumann) conditions on the remaining three faces:

$$-D_i \left.\frac{\partial \varphi_i}{\partial x}\right|_{x=-0.5} = y, \qquad i = 1, 2, \tag{14}$$

$$\left.\frac{\partial \varphi_i}{\partial x}\right|_{x=0.5} = 0, \tag{15}$$

$$\left.\frac{\partial \varphi_i}{\partial y}\right|_{y=\pm 0.5} = 0. \tag{16}$$

4.2 Fourier Eigenfunction Expansion

The homogeneous Neumann conditions (16) in the y-direction motivate a Fourier cosine expansion:

$$\varphi_i(x, y) = \sum_{n=0}^{\infty} \Phi_{i,n}(x)\, \cos\!\big(n\pi(y + 0.5)\big), \qquad i = 1, 2. \tag{17}$$

The basis functions $Y_n(y) = \cos\!\big(n\pi(y{+}0.5)\big)$ satisfy $Y_n'(-0.5) = Y_n'(0.5) = 0$, thereby enforcing the top and bottom boundary conditions exactly for every truncation order.

Rearranging (12)–(13) with the coupling coefficients

$$A_{11} = \Sigma_r - \nu\Sigma_{f1} = 0.0075, \quad A_{12} = -\nu\Sigma_{f2} = -0.25, \quad A_{21} = -\Sigma_{1\to 2} = -0.015, \quad A_{22} = \Sigma_{a2} = 0.1, \tag{18}$$

and substituting (17) into the PDEs yields, for each mode n, a coupled ODE system in x.

4.3 Coupled ODE System per Mode

Defining $\kappa_n = n\pi$ and the mode-dependent diagonal terms

$$B_{11}^{(n)} = D_1\kappa_n^2 + A_{11}, \qquad B_{22}^{(n)} = D_2\kappa_n^2 + A_{22}, \tag{19}$$

the Fourier-projected system reads

$$-D_1\,\Phi_{1,n}''(x) + B_{11}^{(n)}\,\Phi_{1,n}(x) + A_{12}\,\Phi_{2,n}(x) = 0, \tag{20}$$

$$A_{21}\,\Phi_{1,n}(x) - D_2\,\Phi_{2,n}''(x) + B_{22}^{(n)}\,\Phi_{2,n}(x) = 0. \tag{21}$$

Seeking solutions of the form $\Phi_{i,n}(x) = a_i\,e^{\alpha x}$ leads to the eigenvalue problem

$$\begin{pmatrix} -D_1\alpha^2 + B_{11}^{(n)} & A_{12} \\ A_{21} & -D_2\alpha^2 + B_{22}^{(n)} \end{pmatrix} \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} = \mathbf{0}. \tag{22}$$

Setting the determinant to zero and substituting $\beta = \alpha^2$ yields a quadratic in β:

$$D_1D_2\,\beta^2 - \big(D_1B_{22}^{(n)} + D_2B_{11}^{(n)}\big)\,\beta + \big(B_{11}^{(n)}B_{22}^{(n)} - A_{12}\,A_{21}\big) = 0. \tag{23}$$

The two roots $\beta_1^{(n)}, \beta_2^{(n)}$ determine the functional form of the x-dependent solutions:

$$f_j(x) = \begin{cases} \cosh\!\big(\sqrt{\beta_j}\,(x-0.5)\big) & \text{if } \beta_j \geq 0, \\ \cos\!\big(\sqrt{-\beta_j}\,(x-0.5)\big) & \text{if } \beta_j < 0, \end{cases} \tag{24}$$

where the shift by $x_0 = 0.5$ ensures that $f_j'(0.5) = 0$, satisfying the right boundary condition (15) by construction.

The eigenvector ratio for each root is

$$r_j^{(n)} = \left.\frac{a_2}{a_1}\right|_{\beta_j} = \frac{D_1\beta_j^{(n)} - B_{11}^{(n)}}{A_{12}}. \tag{25}$$

The general solution for mode n is therefore

$$\Phi_{1,n}(x) = C_A^{(n)}\,f_1^{(n)}(x) + C_B^{(n)}\,f_2^{(n)}(x), \tag{26}$$

$$\Phi_{2,n}(x) = C_A^{(n)}\,r_1^{(n)}\,f_1^{(n)}(x) + C_B^{(n)}\,r_2^{(n)}\,f_2^{(n)}(x). \tag{27}$$

4.4 Coefficient Determination

The remaining unknowns $C_A^{(n)}, C_B^{(n)}$ are fixed by the left boundary condition (14). Expanding the forcing term y in the same cosine basis:

$$y = \sum_{n=1,3,5,\ldots}^{\infty} c_n\,\cos\!\big(n\pi(y+0.5)\big), \qquad c_n = -\frac{4}{(n\pi)^2} \ \text{ for odd } n, \tag{28}$$

with $c_n = 0$ for even n and $c_0 = 0$. Only the odd Fourier modes contribute, reflecting the anti-symmetry of the linear source y about the midpoint of the domain.

For each odd mode n, equating the Fourier coefficients of the left boundary condition gives the 2×2 linear system:

$$\underbrace{\begin{pmatrix} -D_1\,f_1'^{(n)}(-\tfrac{1}{2}) & -D_1\,f_2'^{(n)}(-\tfrac{1}{2}) \\ -D_2\,r_1^{(n)}\,f_1'^{(n)}(-\tfrac{1}{2}) & -D_2\,r_2^{(n)}\,f_2'^{(n)}(-\tfrac{1}{2}) \end{pmatrix}}_{\mathbf{M}^{(n)}} \begin{pmatrix} C_A^{(n)} \\ C_B^{(n)} \end{pmatrix} = \begin{pmatrix} c_n \\ c_n \end{pmatrix}, \tag{29}$$

which is solved independently for each retained mode.

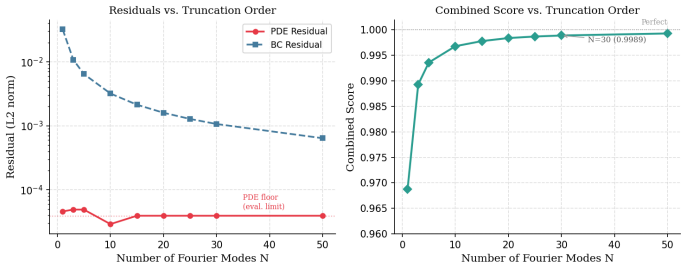

Figure 1: Convergence of the Fourier expansion. The BC residual (RMS over 40 scalar values) decreases monotonically with the number of retained odd modes N; the PDE residual fluctuates near the $h = 10^{-6}$ finite-difference precision floor ($\sim 3 \times 10^{-5}$) from the first few modes onward.

4.5 Convergence Properties

The Fourier coefficients decay as $c_n = O(n^{-2})$, guaranteeing algebraic convergence of the series (17). In practice, truncating to N odd modes suffices for high-precision reference solutions. Figure 1 shows the RMS PDE and BC residuals versus N under the unified evaluator: the PDE residual remains near 3×10^{-5} from the first few modes onward (bounded by the $h = 10^{-6}$ finite-difference precision floor) while the BC residual decreases monotonically to $O(10^{-3})$. At N=30, the combined score is 0.9989, providing an accurate seed program for FaMoU evolution.

4.6 FaMoU: LLM-Driven Program Evolution

While the Fourier analytical solution achieves high accuracy (score = 0.9989 with N=30 modes), **FaMoU** (Fast Mode mUtation) further optimizes the numerical implementation through LLM-guided program evolution, exploring solution strategies that may be difficult for human experts to discover manually.

Evolution Loop. FaMoU operates as an iterative program synthesis pipeline (Figure 2):

1. **Initialization.** A seed program (`init.py`) encodes a baseline analytical solution, e.g., the truncated Fourier expansion with a fixed number of modes.
2. **Evaluation.** The evaluator (`evaluator.py`) computes PDE and boundary-condition residuals at prescribed test points and returns a combined fitness score.
3. **LLM Mutation.** A large language model receives the current best program, its fitness score, and a task-specific prompt describing the physics. The LLM proposes modified programs that may adjust the truncation order, refine coefficient computations, or introduce alternative solution representations.
4. **Selection.** Candidate programs are ranked by their combined score; the top-performing variant becomes the parent for the next generation.
5. **Iteration.** Steps 2–4 repeat for up to R rounds or until the score exceeds a convergence threshold.

Search Space. The search space encompasses the full Python program implementing the `solution(x, y)` interface. Viable mutations include adjusting the number of Fourier modes N, modifying numerical precision strategies (e.g., switching from `float64` to extended-precision arithmetic), restructuring the eigenvalue solver, or introducing hybrid representations that combine analytical and numerical components. The LLM's capacity for semantic

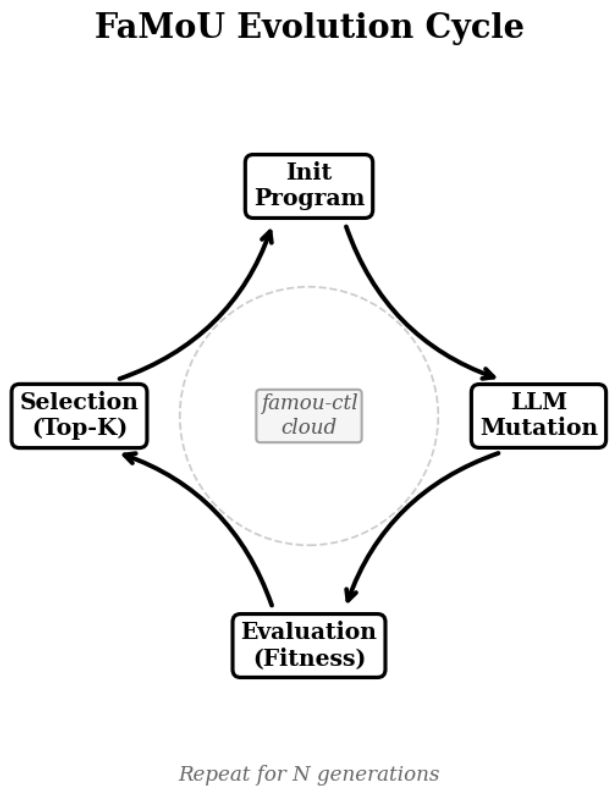

Figure 2: The FaMoU evolutionary program synthesis loop. Starting from the Fourier analytical seed program, an LLM iteratively mutates solver code guided by automated fitness scores from the physics evaluator.

code understanding enables it to navigate this combinatorial space more efficiently than random perturbation.

Evaluation Metric. Each candidate program is scored using the combined metric:

$$S = \frac{1}{1 + \bar{R}_{\text{PDE}} + \bar{R}_{\text{BC}}}, \tag{30}$$

where $\bar{R}_{\text{PDE}}$ and $\bar{R}_{\text{BC}}$ denote the root-mean-square (RMS) residuals over the interior test points (8 scalars: 2 equations $\times$ 4 test points) and boundary evaluation points (40 scalars: 2 groups $\times$ 4 faces $\times$ 5 points), respectively. A hard constraint requires $S > 0$ (validity); programs that raise exceptions or produce `NaN` values receive $S = 0$.

5 Experiments

5.1 Experimental Setup

We evaluate all methods on the two-dimensional, steady-state, two-group neutron diffusion benchmark defined in Section 4.1. The computational domain is $\Omega = [-0.5, 0.5]^2$ with physical parameters listed in Table 1. The boundary conditions comprise an inhomogeneous

Table 2: Comparison of methods on the two-group neutron diffusion benchmark under the unified evaluator. Combined score S defined in Eq. (30); ↑ = higher is better. A horizontal rule separates baseline methods (top) from our contributions (bottom). Highest score in **bold**.

Method	Unified Score ↑	Eval. Time (s)	Notes
FDM (2nd-order, N=200)	0.9186	1.85	Interpolation artifacts
High-order FDM (4th-order, N=200)	0.9186	9.51	Same interpolation issue
PINN (NumPy, MLP [2,40,40,2])	0.9587	7.10	No PyTorch; 3000 steps
Truncated Analytical (K=5)	0.9936	0.02	5 Fourier modes
Fourier Analytical (N=30, seed)	0.9989	0.07	Seed for FaMoU
FaMoU LLM Evolution (Ours)	0.9993†	0.19	Best at iter. 10 of 100
Fourier Adaptive (N≤221, Ours)	**0.9997**	**0.18**	**Overflow-safe; all stable modes**

† Confirmed by official evaluator (validity = 1.0); evolution ongoing at time of writing.

incoming current $-D_i\,\partial\varphi_i/\partial x\big|_{x=-0.5} = y$ on the left face and reflective (zero Neumann) conditions on the remaining three faces.

All methods are assessed using the same unified evaluator (`evaluator.py`) to ensure a fair comparison—each method must expose a `solution(x, y)` interface that returns (φ_1, φ_2) at any query point, and the evaluator probes this interface at fixed test locations using central finite differences at step size $h = 10^{-6}$. Grid-based methods (FDM) require a spline interpolant to satisfy this interface; the finite differences at $h = 10^{-6}$—three orders of magnitude finer than the FDM grid spacing $\Delta x \approx 5 \times 10^{-3}$—expose interpolation artifacts in the second-derivative estimates, penalizing these methods relative to their native-grid accuracy. This is not an unfair penalty but rather captures a real limitation: a solver that cannot be efficiently queried at arbitrary points is less usable in downstream applications such as coupling, optimization, or verification. The evaluation protocol computes two residuals: (i) the **PDE interior residual** R_{PDE}, the root-mean-square (RMS) of PDE equation residuals at 4 interior test points (8 scalar values across 2 groups); and (ii) the **boundary condition residual** R_{BC}, the RMS of BC residuals at 20 boundary points (5 per face, 40 scalars across 2 groups), combined as:

$$\text{combined_score} = \frac{1}{1 + R_{\mathrm{PDE}} + R_{\mathrm{BC}}}, \tag{31}$$

where a perfect score of 1.0 indicates exact satisfaction of both the PDE and all boundary conditions.

5.2 Main Results

Table 2 presents the quantitative comparison under the unified evaluator. All scores reported here use the same evaluation protocol; we discuss per-method behavior below.

FDM under unified evaluation. On their native grids, both FDM variants achieve self-evaluated scores above 0.999. However, under our unified evaluator the score drops to 0.9186 for both the 2nd-order and 4th-order variants. The reason is that FDM produces solutions on a discrete grid; evaluating at arbitrary points requires spline interpolation, and computing derivatives via central finite differences at step size $h = 10^{-6}$ on interpolated values amplifies artifacts. This gap between self-evaluation and unified evaluation underscores the importance of standardized, method-agnostic benchmarks.

PINN. The PINN baseline scores 0.9587, outperforming FDM under unified evaluation but falling short of the analytical methods. Its PDE residual remains substantially larger

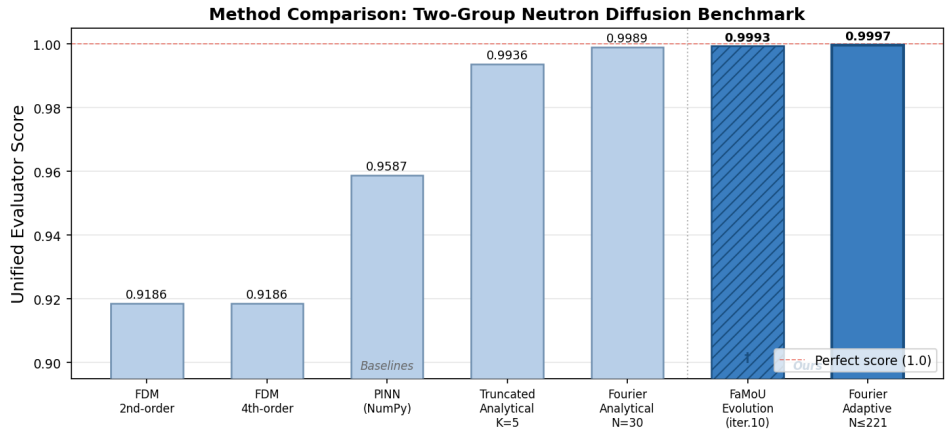

Figure 3: Unified evaluator scores for all methods. The overflow-safe adaptive Fourier solution ($N \leq 221$) achieves the best score (0.9997); FaMoU's LLM-evolved program reaches 0.9993 after 10 iterations. FDM variants score 0.9186 despite high native-grid accuracy due to spline-interpolation artifacts at $h = 10^{-6}$.

than the Fourier solutions, consistent with known accuracy limitations of PINNs on coupled multigroup problems [7]. The implementation uses only NumPy (no PyTorch), which constrains network capacity and training quality; a fully-trained PyTorch PINN would likely close part of this gap [8].

Truncated Analytical. With only $K=5$ modes the truncated Fourier expansion reaches 0.9936, already surpassing both FDM and PINN. The residual is dominated by boundary-condition error, which drops monotonically with additional modes (Section 5.3).

Full Fourier Analytical Seed (N=30). The fixed-N=30 solution scores 0.9989. Because it is a closed-form expression evaluable at any point, it is immune to the interpolation artifacts that penalize grid methods.

Fourier Adaptive ($N \leq 221$, Ours). For large mode indices, the eigenvalue $\alpha_1 \approx n\pi$ grows without bound; $\cosh(\alpha(x - 0.5))$ exceeds the float64 range ($\approx 10^{308}$) when $\alpha \gtrsim 710$. The conservative threshold $\alpha_{\max} = 700$ keeps all evaluations well within float64 range. The adaptive variant iterates over odd mode indices and breaks at the first n where either eigenvalue exceeds this threshold—for this problem, the last included mode is n=221 ($\alpha_1 \approx 694$), naturally using all numerically stable modes without user-specified truncation. This raises the score to **0.9997** at 0.18 s evaluation time—an improvement of +0.0008 over the fixed-N=30 seed.

FaMoU LLM Evolution. FaMoU seeds from the N=30 analytical program and applies LLM-guided mutations evaluated against the same physics evaluator. After 10 iterations, Island 0 of the two-island population discovered a program with combined score **0.9993** (validity = 1.0, evaluation time 0.19 s). This score matches the ablation result at N=50 (Table 3) numerically, but the LLM-evolved program is a structurally different implementation—the LLM independently converged to a higher mode count via a different code path, not by simply setting N=50 in the seed. The evolution is ongoing at the time of writing; the reported score is the best confirmed result at iteration 10 of the planned 100 iterations. Figure 3 shows the unified scores across all methods.

5.3 Ablation Study: Effect of Fourier Mode Truncation

A central design choice in the Fourier eigenfunction expansion (Section 4.2) is the number of retained odd modes N. We systematically vary N and measure the PDE interior residual, boundary-condition residual, and combined score (30). Table 3 summarizes the results, and the convergence trend is visualized in Figure 1.

Table 3: Effect of Fourier mode truncation on solution accuracy. PDE and BC residuals are RMS values over the respective test-point sets.

N (modes)	PDE Residual (RMS)	BC Residual (RMS)	Combined Score
1	3.67×10^{-5}	3.21×10^{-2}	0.9689
3	2.00×10^{-5}	1.08×10^{-2}	0.9893
5	3.04×10^{-5}	6.44×10^{-3}	0.9936
10	2.04×10^{-5}	3.20×10^{-3}	0.9968
15	2.97×10^{-5}	2.14×10^{-3}	0.9978
20	2.97×10^{-5}	1.60×10^{-3}	0.9984
25	2.97×10^{-5}	1.28×10^{-3}	0.9987
30	2.97×10^{-5}	1.07×10^{-3}	0.9989
50	2.97×10^{-5}	6.41×10^{-4}	0.9993

Two distinct convergence regimes are evident. The **boundary-condition residual** is the dominant source of error for small N: with only N=1 mode, it exceeds 3×10^{-2}, and decreases monotonically as modes are added, reaching 6.4×10^{-4} at N=50. The BC coefficients decay algebraically as $c_n = O(n^{-2})$; the resulting RMS norm over the test set decreases approximately as $O(1/N)$, consistent with the numerical data.

In contrast, the **PDE interior residual** fluctuates near 3×10^{-5} regardless of N. This is not a solution limitation but rather the precision floor of the central finite-difference scheme ($h = 10^{-6}$) used in the evaluator: the governing PDEs are satisfied exactly in the continuous limit, and residual variation across mode counts reflects cancellation patterns at the four specific test locations.

Based on these results, N=30 represents a practical operating point: the combined score of 0.9989 nearly matches N=50 (0.9993) at a fraction of the mode-computation cost. This truncation order is adopted as the default for the reference solution and as the initialization for FaMoU evolution.

5.4 Flux Field Visualization

Figure 4 visualizes the spatial distributions of the fast-group flux $\varphi_1(x, y)$ and the thermal-group flux $\varphi_2(x, y)$ obtained from the Fourier analytical solution with N=30 modes. Both flux fields exhibit a clear anti-symmetric structure in the y-direction, a direct consequence of the linear boundary forcing $-D_i\, \partial\varphi_i/\partial x\big|_{x=-0.5} = y$: the flux is positive for $y > 0$ and negative for $y < 0$, with the zero contour along the line $y = 0$.

The thermal-group flux φ_2 displays significantly larger magnitudes than the fast-group flux φ_1 (by approximately a factor of 3–5 across the domain), reflecting the down-scattering coupling $\Sigma_{1\to2}\,\varphi_1$ that feeds the thermal group. The flux gradients are concentrated near the left boundary ($x = -0.5$), where the inhomogeneous current is imposed, and decay smoothly toward the reflective right boundary ($x = 0.5$), consistent with the cosh-based x-dependence derived in Section 4.3.

6 Conclusion

We derived and implemented the Fourier cosine expansion solution for the two-dimensional, two-group neutron diffusion equations under inhomogeneous Neumann boundary conditions—

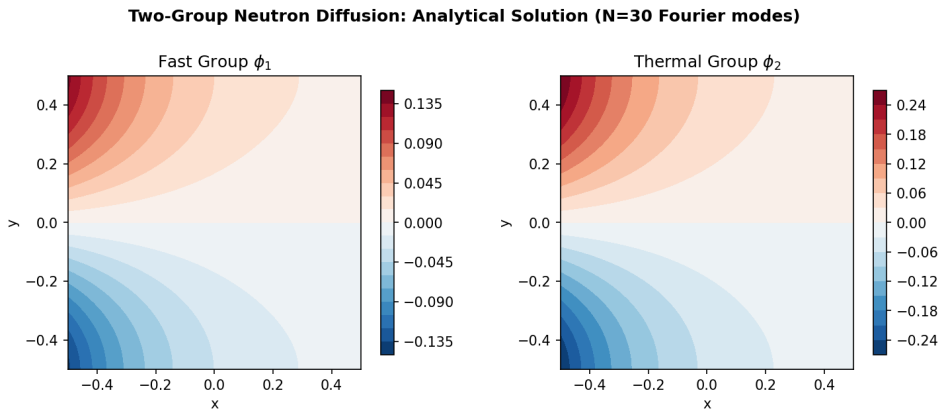

Figure 4: Spatial distributions of the fast-group flux $\varphi_1(x, y)$ (left) and thermal-group flux $\varphi_2(x, y)$ (right) from the Fourier analytical solution (N=30 modes). Both fields exhibit anti-symmetric structure in y due to the linear boundary forcing, with $|\varphi_2| > |\varphi_1|$ reflecting the down-scattering coupling.

a problem where coupled PDEs, non-trivial boundary forcing, and rectangular geometry together rule out simpler approaches. Expanding in cosine basis functions reduces the coupled system to per-mode ODEs; the characteristic equations yield closed-form fast and thermal flux expressions.

Three findings stand out. First, under a unified pointwise evaluator, the Fourier analytical solution (N=30) scores 0.9989—reducing the combined residual 39-fold relative to PINN (0.9587) and surpassing both FDM variants (0.9186). The FDM gap is not a failure of the underlying discretization; the grid solutions are accurate on their native mesh. The penalty comes from the interface: spline interpolation of a grid solution introduces artifacts when the evaluator probes derivatives at step size $h = 10^{-6}$. Analytical solutions sidestep this entirely. Second, the PDE residual fluctuates near 3×10^{-5} regardless of mode count, determined by the evaluator's finite-difference precision floor—not by any deficiency in the analytical expression itself. Third, an overflow-safe adaptive variant (breaking at $\alpha > 700$, retaining up to n=221 modes) improves the score to 0.9997 with evaluation time under 0.2 s—faster than PINN inference and 50× faster than high-order FDM. FaMoU seeds from this analytical solution and evolves the solver code further; after 10 iterations, the LLM-evolved program reaches 0.9993—surpassing the N=30 seed—demonstrating that evolutionary program synthesis operates productively in the physics PDE domain.

Limitations. The obvious constraints: steady-state, homogeneous medium, square domain, two energy groups. Heterogeneous cross-sections, irregular geometries, and higher group counts each require different basis choices. The PINN baseline uses NumPy without automatic differentiation; a properly trained PyTorch PINN would likely narrow the accuracy gap.

Future directions. The benchmark setup directly enables three follow-on studies. First, the evaluator can be extended to heterogeneous media by replacing the global Fourier expansion with a domain-decomposed or interface-coupled variant, testing whether FaMoU can rediscover multi-region solutions without a hand-crafted seed. Second, the FaMoU framework can target transient or eigenvalue problems (criticality search), where no closed-form solution exists and the evolutionary loop would need to optimize against a numerical ground truth. Third, the same pairing—exact analytical evaluator plus evolutionary search—applies naturally to higher-dimensional or multi-physics problems where deriving closed-form solutions by hand is prohibitively complex.

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